D07 · Warehouse Automation & AMR/Robotics

03 · Pick-to-light, put-to-light & voice-directed picking

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1 WHAT IT IS

Assisted picking technologies guide the operator to the right item and quantity: pick-to-light and put-to-light use lights at locations, voice picking gives spoken instructions, and RF scanners verify each pick by barcode.

2 WHY IT MATTERS

These technologies mainly reduce errors and secondarily raise productivity, without changing the warehouse layout. **Their accuracy depends on real confirmation: an operator who confirms without checking defeats any system.** Each fits a different profile — lights for dense, fast, fixed locations; voice for hands-free picking in mid-to-high volume; scanners as the universal baseline.

3 HOW IT IS DONE

1. **Measure the error baseline by zone.** Picking errors and their cost — correction, reshipping, customer impact.
2. **Match technology to the zone.** Pick-to-light for fast fixed-location areas, voice or RF elsewhere.
3. **Integrate confirmations with the WMS.** Every pick recorded automatically, no extra step.
4. **Audit by operator and zone.** Find where accuracy falls and why; retrain or redesign.

IN PRACTICE

Voice-directed picking is the highest ROI assisted picking technology for most warehouses

More flexible than pick-to-light, with similar accuracy (>99.9%), and improves productivity 15–30% vs. RF scanner.

Pick-to-light is the correct technology for high-speed zones with fixed SKUs

In zones where the same SKUs are picked with very high frequency and locations are fixed, pick-to-light is faster than voice-directed.

Integrate assisted picking technology with the WMS for automatic confirmation

Pick confirmation must be recorded automatically in the WMS — without an additional operator action.

Measure pick accuracy by technology, zone, and operator to identify actionable problems

DC average accuracy can hide that a specific zone or operator has significantly lower accuracy.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: Voice-Directed Picking — pharmaceutical distribution DC, 12,000 lines/day — Migration from RF scanner to voice-directed picking.

Metric	RF Scanner (before)	Voice-Directed Picking (after)
Pick accuracy %	MEDIUM	99.2% — 58 errors/day at 12,000 lines
Pick accuracy % with voice	OK	99.97% — 4 errors/day (-93% errors)
LPPH (lines per hour)	OK	68 LPPH -> 84 LPPH (+24%)

Result: Voice-directed picking ROI at 18 months: $480K in error correction cost reduction + $220K in additional productivity vs. $180K voice system investment.

HOW IT IS MEASURED

Directed Pick Accuracy %

Directed Pick Accuracy %(Lines executed correctly with the technology / Total technology-assisted lines) × 100

Benchmark: >99.9% for pick-to-light and voice-directed · >99.5% for RF scanner Directed pick accuracy <99.5% with voice-directed indicates operators are confirming locations without physically verifying the product — a process problem the system alone cannot solve.

Voice-Directed LPPH vs. RF Scanner LPPH

Voice-Directed LPPH vs. RF Scanner LPPHLPPH with voice-directed / LPPH with RF scanner

Benchmark: 15–30% LPPH improvement migrating from RF scanner to voice-directed LPPH improvement comes primarily from eliminating screen reading time and manual confirmation. The voice operator has hands and eyes free for picking.

WHAT YOU WOULD USE

Pick Assist Technology

Zebra Technologies / Honeywell Vocollect — Module: RF Scanners + Voice-Directed Picking HardwareThe reference assisted picking hardware providers. Honeywell Vocollect leads in voice-directed; Zebra leads in RF scanners.

Knapp / Dematic — Module: Pick-to-Light + Put-to-Light SystemsThe providers of pick-to-light and put-to-light systems for high-speed warehouses.

4 SO WHAT

Guide the picker, then verify the pick. Baseline errors, choose lights, voice or RF by zone, integrate with the WMS and audit where accuracy slips.

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